

Light and Dark Mode Changes

Project: Monash University static website

Files changed: monash.html and monash.css

This document explains the changes added to support both light mode and dark mode. The update keeps the existing page structure, adds a working theme toggle, saves the selected theme, and updates page colors through CSS variables.

Area	What changed	Why it matters
HTML	Added a small theme initializer in the head.	Prevents the page from flashing the wrong theme when it loads.
HTML	Updated the existing theme button with text and icon behavior.	Users can switch between Dark and Light from the header.
HTML	Added JavaScript before the closing body tag.	The button updates the page theme, icon, label, aria-label, and saved preference.
CSS	Added color variables for light and dark modes.	The whole page can change theme from one consistent color system.
CSS	Replaced many hardcoded colors with variables.	Cards, banners, news, links, search, nav, and footer respond to the selected theme.
CSS	Added dark-mode logo and surface adjustments.	The header logo and content remain readable in dark mode.

HTML Changes

1. Theme setup script in the head

A script was added before the CSS file loads. It checks `localStorage` for a saved choice named `monash-theme`. If no saved choice exists, it checks the user's system preference with `prefers-color-scheme`. Then it sets `document.documentElement.dataset.theme` to either `light` or `dark`.

Key idea: the `html` element receives `data-theme="light"` or `data-theme="dark"` before the stylesheet is applied.

2. Theme button label

The existing button with class `theme-toggle` already existed in the header. I added a `span` next to the icon so the button can show `Dark` in light mode and `Light` in dark mode. The `aria-label` also changes so screen readers understand the action.

3. Toggle script before closing body

A small JavaScript block finds the theme button, updates the icon and label, switches the `data-theme` value when clicked, and saves the user's choice in `localStorage`. This means the selected theme stays active after refreshing the page.

CSS Changes

1. CSS variables

At the top of monash.css, I added a :root block for light-mode colors. These variables include page background, surface color, text color, muted text, link color, brand blue, borders, icon background, and footer colors.

Examples: --page-bg, --surface, --text, --muted-text, --link, --brand-blue, --border.

2. Dark-mode variable override

A [data-theme="dark"] block overrides the same variables with dark colors. Because the page uses variables, changing one data attribute on the html element updates the theme across the site.

3. Updated page sections

Hardcoded colors were replaced with variables across the body, headers, dropdown menu, search field, intro text, info cards, banners, buttons, news cards, links, and footer. This keeps contrast readable in both themes.

4. Theme toggle styling

The theme toggle was styled as a compact pill button with an icon and text. It has hover and focus-visible styles so it feels interactive and remains keyboard accessible.

5. Dark logo support

The Monash logo in the second header was filtered in dark mode so it stays visible on the dark header surface.

6. News section support

The news cards were given themed background, border, text, link, and muted text colors so this section does not remain partially light or unreadable in dark mode.

How The Theme Works

The implementation uses one simple rule: JavaScript changes the data-theme attribute, and CSS reads that attribute to choose colors.

Step	Action
1	Page loads and checks saved theme.
2	If nothing is saved, it checks system preference.
3	The html element gets data-theme set to light or dark.
4	CSS variables apply the matching color palette.
5	When the user clicks the button, JavaScript switches the theme and saves it.

Testing Performed

The page was served locally at <http://127.0.0.1:8087/monash.html> and opened in the browser. I checked the initial light theme and clicked the theme toggle. After clicking, the page changed to dark mode, the body background changed to a dark color, the card surfaces changed to dark surfaces, and the button label changed from Dark to Light.

Confirmed dark-mode values included: body background `rgb(16, 21, 27)`, body text `rgb(243, 247, 251)`, card surface `rgb(22, 29, 37)`, and aria-label 'Switch to light mode'.

Files Affected

monash.html: added the theme initializer, button label, and click behavior.

monash.css: added theme variables and updated styles to use them across the page.

Generated for the Monash static website theme update.